

NEET Important Chapter – Unit and Measurements

Important Concept of Unit and Measurements for NEET

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In this chapter, we will get an overall understanding of studying units and measurements. This chapter deals with the basics of physics. In this chapter, we will get to know about physical quantities and their units. Physical quantities are divided into two major categories. One is fundamental physical quantities and the second one is derived physical quantities. Similarly, units are also categorized as fundamental or standard units and derived units. This chapter deals with these physical quantities and their units. In this chapter, we will get to know how to define a unit for any physical quantity.

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The quantity measurement consists of two parts. The first part shows the frequency of the standard unit of measurement and the second part shows the name of the unit. This chapter describes the fundamental units of fundamental physical quantities such as length, time, mass, and temperature, and the derived quantities expressed in standard units of measurement. It may be a quantity. You can do a dimensional analysis of the size and then check if the formula of any size is correct. Suppose you measure something and the measurement causes an error. After learning this chapter, we can find the error in the measurement. We can learn how to use the basic quantity to find the dimension of the derived physical quantity.

Let's discuss the important concepts and formulae related to NEET physics units and measurement exams along with a few solved examples.

Important Topics of Unit and Measurements

- Physical Quantities
- Fundamental and derived quantities
- System of units
- Practical Units
- Dimensions of physical quantities
- Dimensionless quantities
- Applications of dimensional analysis
- Significant figures
- Errors of measurement

Some Important Concept of Unit and Measurements for NEET

Name of the Concept	Key Points of the Concepts
1. Physical Quantities	A physical quantity is known as the physical property of a material or system that can be quantified by measurement. Physical quantities can be expressed as values that are algebraic multiplications of numbers and units.

2. Types of Physical Quantities and Units	Physical quantities are classified into two parts. • Fundamental Physical Quantities or Standard physical quantities are measured in the basic units of measurement. These are seven fundamental physical quantities, measured in base or physical fundamental units: length, mass, time, electric current temperature, amount of substance, and luminous intensity. • Derived physical quantities are calculated by two or more measurements including volume, density, and area. Whereas, the area of a rectangular surface is calculated as its length multiplied by its width. The volume of a rectangular solid is calculated by the product of length, width, and height.
3. Dimension of Physical Quantity	The dimension of a physical quantity may be defined as the number of times the fundamental units of mass, length and time appear in the physical quantity.
4. Dimensionless Physical Quantities	Dimensionless quantity is also known as the quantity of dimension with one as a quantity that is not related to any physical dimension.
5. Applications of Dimensional Analysis	There are many applications where Dimensional analysis plays an important role. Some applications are listed below- • Finding units of a physical quantity in a given system of units. • Finding dimensions of physical constant or coefficients. • Conversion of physical quantities from one system to another. • Checking the dimensional correctness of any given physical relation.

6. Significant Figures	The term "significant digit" refers to the number of significant digits (0 to 9) in the representation factor of scientific notation. The number of significant digits in an expression indicates the certainty or accuracy with which an engineer or scientist expresses the quantity.
7. Error In Measurements	In Measurements, there are two types of errors in physics- • Random Errors • Systematic Errors Random Errors: These errors occur due to some unknown reasons. We can't eliminate it completely but we can reduce it by taking several readings of the same quantity and then taking their mean value. Systematic Error: These errors occur when all the measurements of physical quantities are affected equally, these give the consistent difference in the readings. These errors might occur due to - • Zero errors in measuring instruments • Poor calibration of the instruments • Incorrect calibration on the measuring instruments.

List of Important Formulas of Unit and Measurements

S.No.	Name of the Concept	Formula
1.	Absolute Error for reading	
2.	Mean absolute error	_____
3.	Relative error	_____
4.	Percentage error	_____
5.	Absolute error in sum	• Sum: • Absolute Error:
6.	Absolute error in difference	• Sum: • Absolute Error:
7.	Relative error in product	• Product: • Relative Error: _____
8.	Relative error in division	• Division: _____ • Relative Error: _____

Solved Examples of Unit and Measurements

1. If P, Q, R are physical quantities, having different dimensions, which of the following combinations can never be a meaningful quantity?

(a) _____

(b) _____

(b)

(b) _____

Sol:

Given that,

P, Q, R are three different physical quantities.

To find : Invalid quantity among the options given.

Different physical quantities can not be added or subtracted. As if we add or subtract a physical quantity then units will create the conflict. So from the option we can choose the option which is having direct addition or subtraction of physical quantities.

In option (a), _____ which is not valid so option (a) is not valid.

In option (b), (c), and (d), it is possible to multiply and divide the quantities with each other. So it is not violating the rule of mathematical operation on physical quantities. So it is valid.

Hence, the correct answer is option (a).

Key Point: Validation of any derived unknown physical quantity can be done with the rules of mathematical operation on physical quantities. If it is violating the rule, it can't be a valid physical quantity.

2. A physical quantity X is related to four measurable quantities a, b, c and d as follows: _____ . The percentage error in the measurement of a, b, c and d are 1%, 2%, 6% and 4%, respectively. What is the percentage error in quantity X?

Sol:

Given that,

Percentage error in

Percentage error in

Percentage error in

Percentage error in

To find : Percentage error in

Percentage error in is given by-

$$\frac{\text{Error in } x}{x} \times 100 = \frac{\text{Error in } y}{y} \times 100$$

Here, —

, —

, —

, —

By putting all the value together in the above equation-

$$\frac{\text{Error in } x}{x} \times 100 = \frac{\text{Error in } y}{y} \times 100$$

$$\frac{\text{Error in } x}{x} \times 100 = \frac{\text{Error in } y}{y} \times 100$$

$$\frac{\text{Error in } x}{x} \times 100 = \frac{\text{Error in } y}{y} \times 100$$

Hence, the percentage error in is .

Key Point: Percentage error in any quantity can be calculated by using the standard formula of percentage error.

Previous Year Questions of Unit and Measurements from NEET Paper

1. The unit of thermal conductivity is ()

a.

b.

c.

d.

Sol:

To Find: Unit of thermal conductivity

Formula Used: ————

To get the unit of thermal conductivity we have to go with the definition of thermal conductivity which gives us the formula in which thermal conductivity is present and using that we can get the formula for thermal conductivity.

We know that, the rate of heat flow through a conductor of length l and area of cross-section A is given by -

$$Q = \frac{kA\Delta T}{l}$$

Where, k = Coefficient of thermal conductivity,

ΔT = Change in temperature

By rearranging the equation to get the formula for thermal conductivity, we get-

$$k = \frac{Ql}{A\Delta T}$$

Units of quantities which are on the right side of the equation are-

$$l = \text{Length} = \text{m}$$

$$A = \text{Area} = \text{m}^2$$

$$\Delta T = \text{Temperature Difference} = \text{K}$$

$$Q = \text{Rate of heat flow} = \frac{\text{J}}{\text{s}}$$

Putting all these units together in above equation-

$$k = \frac{\text{J}}{\text{s} \cdot \text{m}^2 \cdot \text{K}} \cdot \text{m} = \frac{\text{J}}{\text{s} \cdot \text{m} \cdot \text{K}}$$

So, option (C) is the correct answer.

Key Point: To get the unit of thermal conductivity we must know the formula of rate of heat flow where thermal conductivity comes in the role. By using that formula and with a little rearrangement in it we can get the unit of thermal conductivity.

2. The unit of permittivity of free space, is ()

a.

b.

c.

d.

Sol:

To Find: Unit of permittivity of free space

Formula Used: _____

To get the unit of permittivity of free space we have to use Coulomb's law of electrostatic force. In Coulomb's law, when there is no medium between two charges it used permittivity of free space for electrostatic force calculations.

By Coulomb's law of electrostatic force-

Here, _____ = charges,

_____ = distance between two charges ,

_____ permittivity of free space.

After rearranging the formula to get the permittivity of free space, we get-

Here, units of

and (Charges) = coulomb (),

(distance) = meter (),

and (Force) = newton ()

If we put all these units together in the above equation, we get-

$$\frac{\text{C}^2}{\text{N} \cdot \text{m}^2} \times \frac{\text{N} \cdot \text{m}^2}{\text{C}^2} = 1$$

Hence the correct option is (C).

Key Point: To solve this question and get the unit or permittivity of free space we must have the knowledge about the Coulomb's electrostatic force.

3. A screw gauge gives the following readings when used to measure the diameter of a wire.

Main scale reading: 0mm

Circular scale reading: 52mm

Given that, 1mm on the main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is

- a. cm
- b. cm
- c. cm
- d. cm

Sol:

Given that,

Main scale reading = MSR = 0mm

Circular scale reading = CSR = 52mm

To Find: The diameter of the wire

Formula Used:

Here, D = Diameter of the wire,

MSR = Main Scale Reading

CSR = Circular Scale Reading

LC = Least Count

First of all we have to calculate the Least Count of the scale. Least count of screw gauge is calculated as

$$\frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$

Now using the formula we can put all the values in the above formula to get the diameter of the wire.

Hence, the option (d) is correct.

Key Point: In this type of question we need the Least count of the measurement device which is defined as the smallest value which can be measured through that device. After calculating the Least count we can use the direct formula to get the answer.

Practice Questions

1. If the L, Q, R represents inductance, charge and resistance respectively, the unit of-

(a) — will be those of current

(b) —

will be those of energy

(c) — will be those of current

(d) — will be those of power

(Ans. option a)

2. A calorie is a unit of heat (Energy in transit) and it equals around where . Suppose we employ a system of units in which the unit of mass equals , the length equals , the unit of time is . Then what will be the magnitude of calories in terms of new units?

(Ans.)

Conclusion

In this chapter we discussed units and measurements. We get to know that in measurement every unit which is not fundamental, can be represented as a fundamental unit. In a measurement device, the least count is the ability of the device to measure the smallest possible value. We also discussed errors in measurement. We went through some formula based problems which are very useful while we are solving the questions in the [NEET exam](#).

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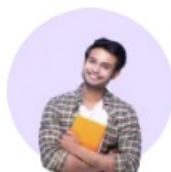
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